

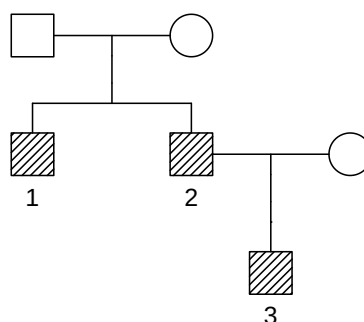
Pedigree analysis and relatedness inference

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Solutions for exercise set III: Relationship inference and pedigree reconstruction

Solution III-1

- The estimates indicate that 2 and 3 are parent-child (in some order), and that 1 and 2 most likely are full siblings. The relationship between 1 and 3 is more distant, compatible both with half siblings (or similar) and first cousins (or similar).
- Assume that 1 and 2 are full siblings. Then, since 3 has a father/son-relationship with 2, but *not* with 1, the only possibility is that 3 is the son of 2. This implies that 3 is the nephew of 1, which is compatible with their point in the triangle above. Thus we are left with this pedigree:



Solution III-2

- 24 markers, as reported by `summary()`:

```
x = readPed("data/reconstruct-fathers.ped")
x = setFreqDatabase(x, "data/reconstruct-fathers.freq")
```

```
summary(x)
```

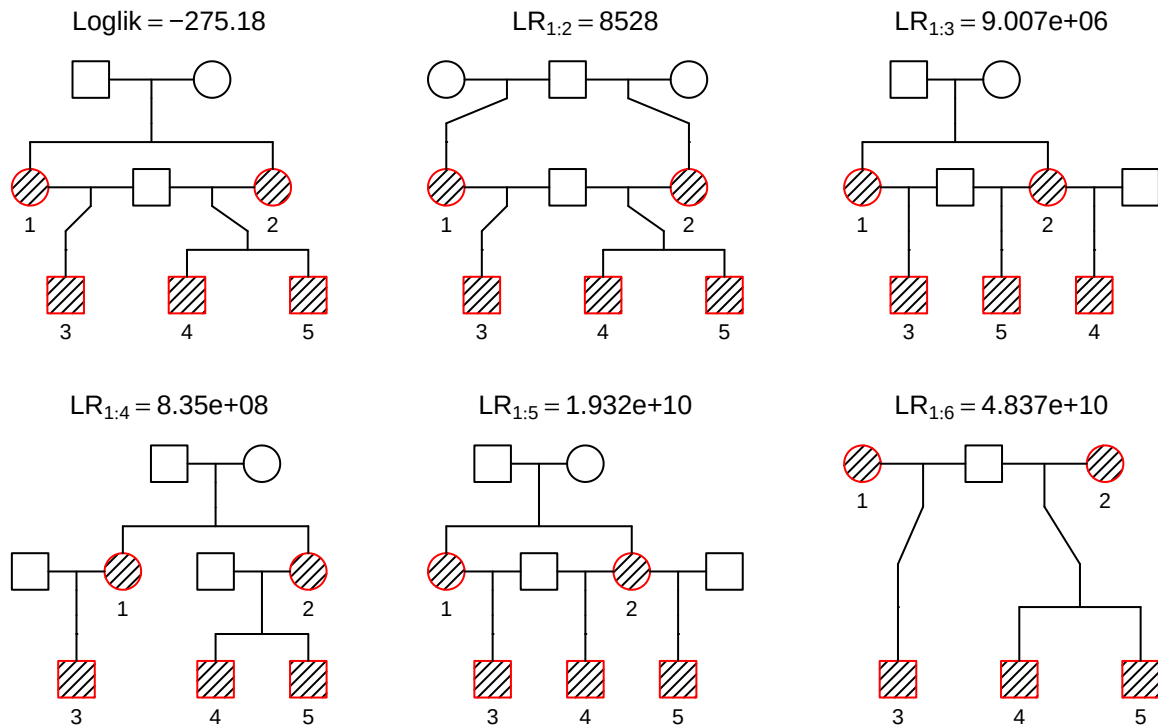
```
## List of 5 singletons.
## Labels: 1 (female), 2 (female), 3 (male), 4 (male), 5 (male).
## 24 attached markers.
```

- Here is one way to do it:

```
r = reconstruct(x, noChildren = 3:5, knownPO = list(c(1,3), c(2,4), c(2,5)))
```

- All three children have the same father! Indeed, this is supported by both the two most likely pedigrees:

```
plot(r, top = 6)
```

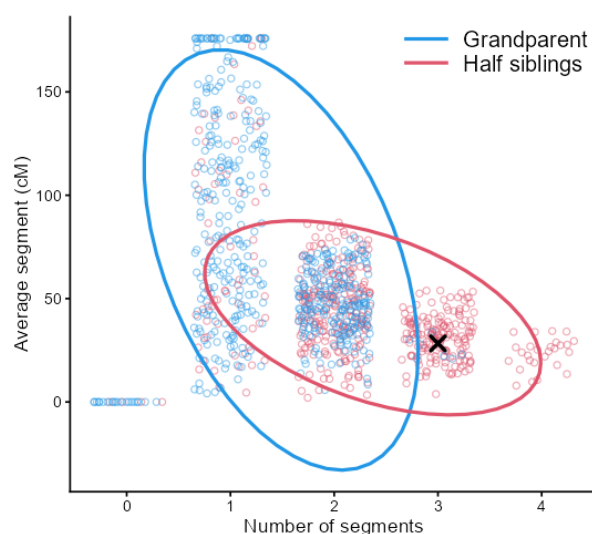


Solution III-3

- a) Paternal half siblings can be ruled out: If either of the individuals are male, their X-chromosome is maternally inherited, so no IBD sharing on X can go through the father. If both individuals are female, they would share the entire paternal X-chromosome, not only three segments.

Paternal grandparent-grandchild can also be disregarded. Again, depending on their sex, they would share either nothing or an entire X chromosome - neither of which is compatible with the observed data.

- b) Maternal half siblings is the best fit. The picture below shows X-chromosomal simulations by **ibdsim2**, with the observed segments represented by the black cross.



Solution III-4

The woman's father is the most likely father of the child. This is supported by the **ibdsim2** simulations below, comparing the distributions of autozygous segments for **Father-daughter incest** vs. **Full-sib**

`incest` (both available in the list of built-in pedigrees).

